



SARDAR PATEL UNIVERSITY
M.Sc. IT (Integrated) NC Exam Oct 2021
7th Semester (CBCS)
Friday, Date: 22/10/2021

Session: Evening

Time: 02:00 PM to 4:00 PM

Course Code: PS07EIT01

Course Title: Image Processing

Total Marks: 70

Q1. A. Multiple Choice Questions. (Attempt All)**[08]**

1. The process of extracting information from the image is called as _____.
 A. Image Segmentation C. Image Enhancement
 B. Image Restoration D. Image Analysis
2. Which is the image processing technique used to improve the quality of image for human viewing?
 A. Enhancement C. Compression
 B. Restoration D. Analysis
3. Thresholding function in contrast stretching creates _____.
 A. Binary image C. High quality image
 B. Poor quality image D. Enhanced image
4. Image can be sharpened using _____.
 A. Low Pass filter C. Contouring
 B. High Pass filter D. Erosion
5. Square of standard deviation is called _____.
 A. Noise C. Power
 B. Spike D. Variance
6. Fourier spectrum of noises are constant and usually called ____ noise.
 A. White C. Black
 B. Red D. Green
7. Histogram equalization refers to image _____.
 A. Sampling C. Filtering
 B. Framing D. Normalization
8. Morphology refers to _____.
 A. Pixel C. Shape
 B. Matrix D. Color

Q1.(B) Fill in the blanks and True/ False. (One mark each)**[16]**

1. The smallest element of an image is called pixel. (True/False)
2. Intensity levels in 8bit image are 250. (True/False)
3. 2D Fourier transform and its inverse are infinitely linear. (True/False)
4. Continuous functions are sampled to form a digital image. (True/False)
5. The purpose of restoration is to gain the original image. (True/False)
6. Gaussian noise is referred to as Gamma noise. (True/False)
7. Information per source is called sampling. (True/False)
8. In the image of dimension $M \times N$, M refers to number of columns ?
 (True/False)

P.T.O.

9. First fundamental step in image processing is _____.
10. Digital images are displayed as a discrete set if _____ values.
11. The product of two even or two odd functions is _____.
12. Fourier transform of unit impulse at origin is _____.
13. Function having both properties of additivity and homogeneity is called _____.
14. Salt and pepper noise also referred to the term _____ noise.
15. Image with very poor quality is considered as _____ image.
16. Error of the image is referred to as _____.

**Q2. Answer the following short questions.
(Attempt any SEVEN OUT OF NINE)**

[14]

1. What is Digital Image Processing?
2. List out first five fundamental steps of DIP.
3. Write applications of image transform.
4. What do you mean by smoothing?
5. Explain various source of noise.
6. What is periodic noise?
7. What are the elements of a digital image?
8. List out basic data redundancies in image compression.
9. What is Data Compression?

Q3. Explain fundamental steps in image processing.

[08]

OR

Q3. Explain components of an image processing system.

[08]

Q4. Explain properties of the 2-D Discrete Fourier Transform.

[08]

OR

Q4. Explain Image Sharpening techniques in detail.

[08]

Q5. Explain a Model of the Image Degradation / Restoration process.

[08]

OR

Q5. Explain Noise Probability Density Functions.

[08]

Q6. Explain source encoder & decoder.

[08]

OR

Q6. Explain variable length coding.

[08]

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